Chapter 2- Software Process Models

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Reference Books:
1) Fundamentals of Software Engineering, Rajib Mall , PHI, Latest edition
2) Software Engineering, I. Sommerville, Pearson Education, Asia.

Topics covered

- Iterative Waterfall model
- Drawbacks of Waterfall model
- V model
- Prototyping model

Assumptions of Classic Waterfall Model

- Too idealistic
  - Assumes that no defect is introduced during any development activity.
  - Assumes that there is no change in the user requirements.

Iterative Waterfall Model

Iterative Waterfall Model

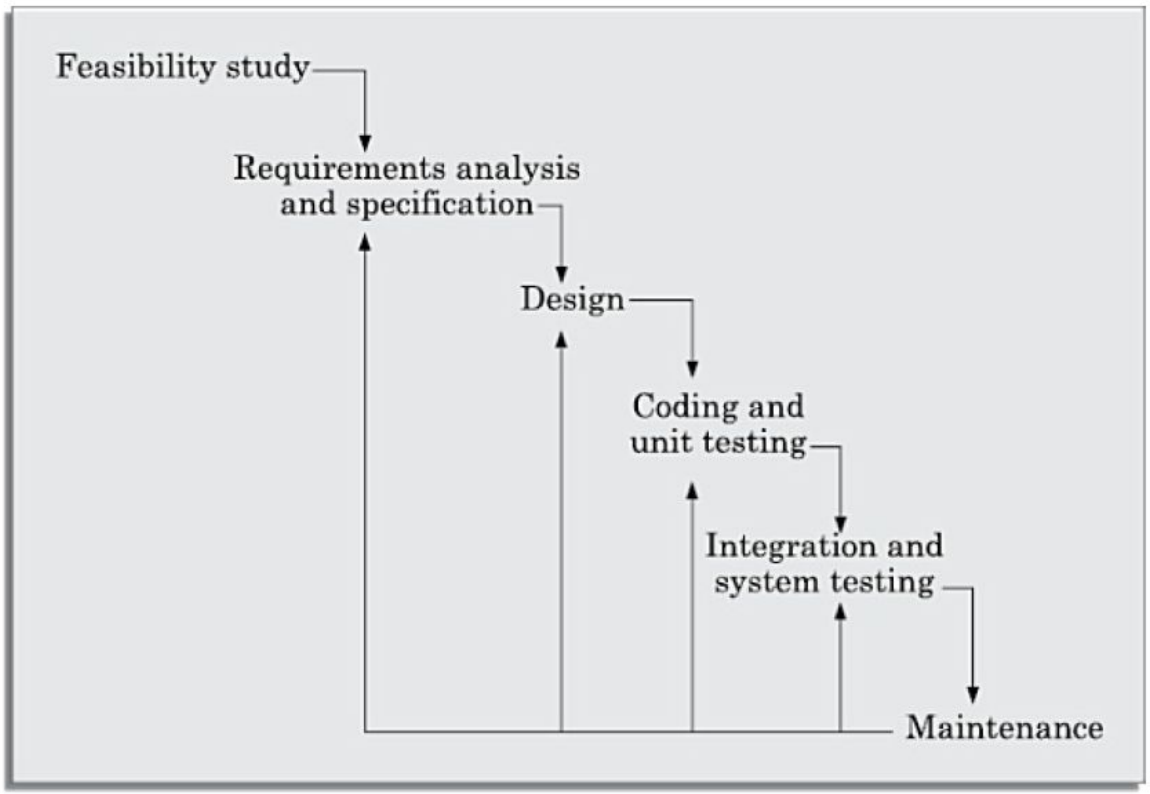

Figure 2.6. Iterative Waterfall model

Iterative Waterfall Model

- Feedback paths are introduced in the classical Waterfall model.
- The defects get detected much later in the life cycle.
- **The later the phase where the defect gets detected, the cost of removing the defect is higher.**

Phase Containment of Errors

- Late detection of error costs time and effort in the development phase
- Efforts should be made that the errors are detected in the same phase that the occur
- Example, detection of a design error during integration testing, then changes need to be made in the ‘design’ and the subsequent phases.

Advantages of Waterfall Model

- The most basic software process model, easily understandable.
- Provides requirements stability
- Milestones are well-understood by team
- Facilitates strong management control

When use Waterfall Model

- Requirements are well-known and stable.
- Technology used is understood.
- An experienced team is developing the project.

- The documentation in any software process model follows the classical waterfall model.

Drawbacks of Waterfall Model

- All requirements should be known upfront
- Can give a false impression of progress
- Expensive error correction – Integration at the end can lead to detection of large number of errors
- Limited customer interaction – Little opportunity for the customer to pre-view the system
- Phase overlap not supported
- No support for risk handling and code re-use

V-model

V Model

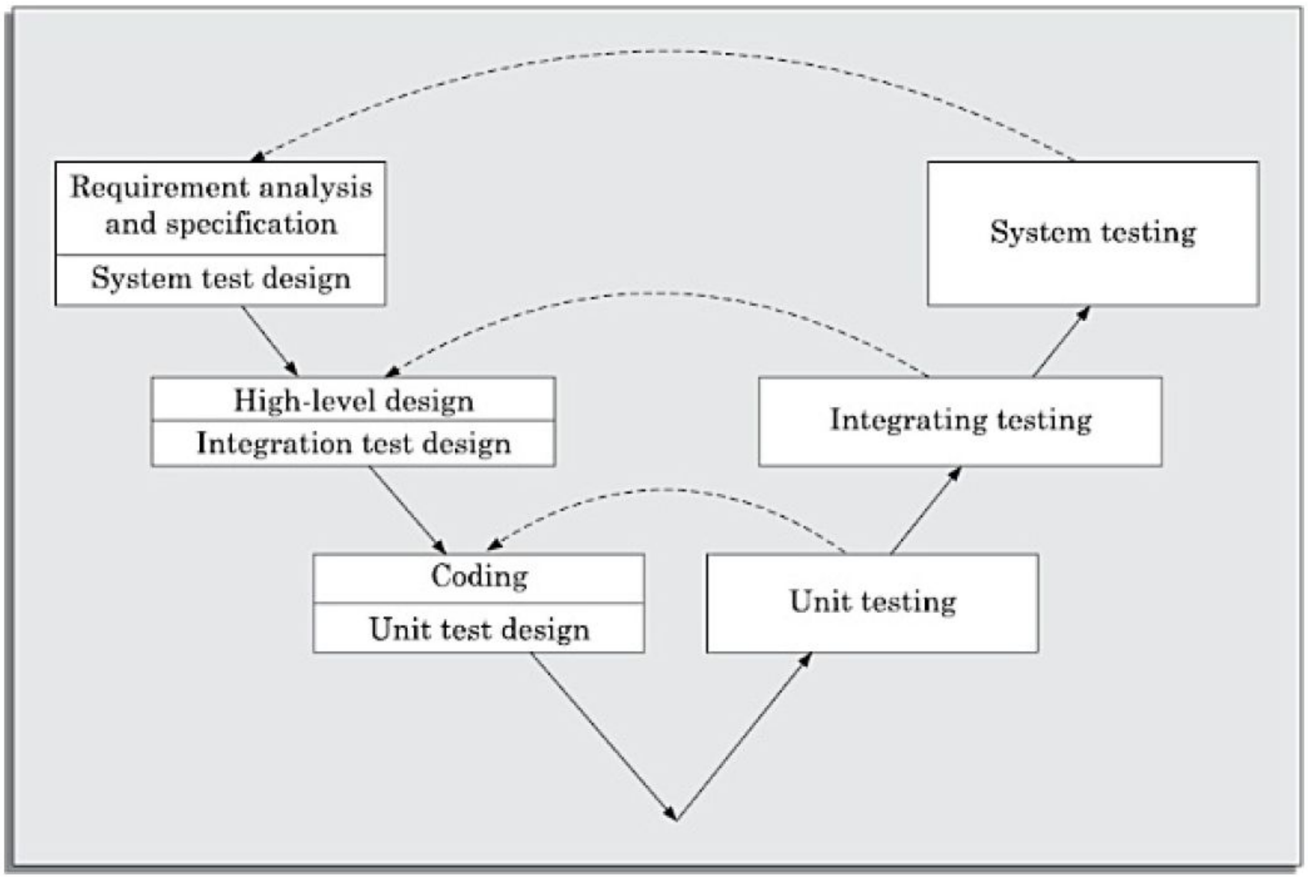

Figure 2.7. V-model

V Model

- Focuses on **Verification and Validation** activities.
- Testing activities are in parallel with every phase of the development.

Weakness of V Model

- Inherits most of the disadvantages of Waterfall model.
- All requirements should be known upfront
- Does not support overlapping between phases
- Does not provide support for risk handling

When to use V Model

- Systems that require high reliability.
- Requirements are well-known and stable.
- Technology used is understood.

Prototyping Model

Prototyping Model

- Before starting the actual development, a prototype of the system is made
- Prototype is a toy implementation of the system.
  - Limited functional capabilities
  - Low reliability
  - Inefficient performance

Prototyping Model

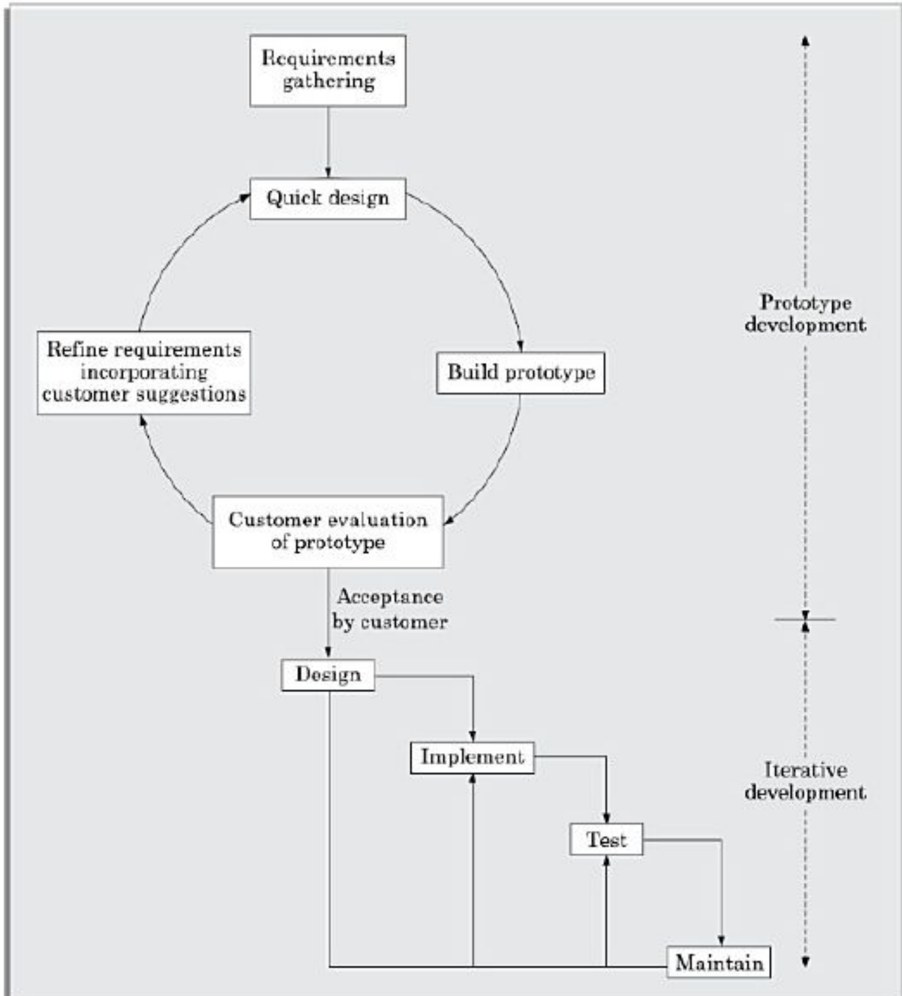

Figure 2.8. Prototyping model

Prototyping Model

- Learning by doing
- Improved communication
- Improved user involvement
- Reduced need for documentation
- Reduced maintenance cost
- Examine technical issues associated with product development

Prototype Construction

- Start with approximate requirements
- Carry out a quick design
- Prototype is built using several short-cuts such as dummy functions, a table look-up rather than performing the actual computations

Prototype Model

- Requirement analysis and specification becomes redundant
  - Final prototype serves as an animated requirement specification
- Design and Code for the prototype are often not used
  - The experience obtained helps in the actual model development

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- Requirement analysis and specification becomes redundant
  - Final prototype serves as an animated requirement specification
- Design and Code for the prototype are often not used
  - The experience obtained helps in the actual model development
- **Apparent cost and actual cost**

Advantages of Prototype Model

- The resulting software is more usable and of better quality
- User needs are better accommodated
- The maintenance cost of the final product is less
- Overall cost is reduced

Disadvantages of Prototype Model

- It may be expensive for some cases
- Susceptible to over-engineering

Next Session

- Incremental Model